

Periodization & Load Management Cheat Sheet

Foundational Principles | PT Student Reference

1. The Four Training Periods (NSCA Framework)

Per NSCA's *Essentials of Strength Training and Conditioning* (4th ed.), the training year splits into four phases, each with distinct objectives:

Phase	Primary Goal	RT Volume (sets/wk, main lifts)	RT Intensity (%1RM)	Frequency
Off-season (Preparatory)	Build base conditioning; increase tolerance for harder training later	High: 12–20 sets/muscle group/wk	65–80%	3–5x/wk
Preseason	Progress from general to sport-specific; convert base to power	Moderate: 9–15 sets/wk	75–90%	3–4x/wk
In-season (Competition)	Maintain strength/power; prioritize recovery and tactical work	Low: 4–8 sets/wk	80–95% (heavy, low volume)	1–2x/wk
Postseason (Active rest)	Recover physically and psychologically; light unstructured activity	Minimal/unstructured	Low, if any	0–2x/wk

(Ranges are general NSCA-consistent guidelines for team-sport athletes — always adjusted for sport, position, and individual training age.)

Off-season = the only period with no competition/tactical demands — this is your window for base-building and higher training volumes.

In-season reality: volume and frequency of lifting typically drop as tactical/technical work takes priority, but intensity is often maintained or even increased with low volume (e.g., heavy, low-rep work 1x/week) to preserve strength and power without adding fatigue.

2. Off-Season Programming Priorities

- **Base hypertrophy and general strength** — higher volume, moderate intensity
 - **Movement quality and correctives** — address asymmetries found in return-to-play testing or prior injury history
 - **Progressive overload is uncontested** — no game-day fatigue to manage around
 - **Testing windows** — beginning and end of off-season for baseline and progress data (strength, jump, sprint, hop testing)
 - **Aerobic base work** — lower priority sports still build a conditioning floor here
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3. In-Season Programming Priorities

- **Maintenance, not development** — the goal is to preserve off-season gains, not chase new PRs
 - **Low volume, moderate-to-high intensity** — 1-2x/week full-body or upper/lower split is common at the college level
 - **Plyometrics at moderate volume** — used to maintain power and rate of force development without excessive soft-tissue load
 - **Autoregulation matters more** — RPE-based load adjustments around game schedule, travel, and practice load
 - **Tactical/technical work takes the training-time priority** — S&C has to fit around it, not compete with it
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4. Rep Ranges by Desired Adaptation

Each adaptation lives in its own %1RM and rep zone. This is the starting point for any exercise prescription decision.

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Adaptation	%1RM	Rep Range	Sets	Rest	Notes
Max Strength	85–100%	1–5	3–5	3–5 min	Near-max loads, full recovery between sets
Power	30–85% (varies by method)	1–5	3–5	2–5 min	Emphasis on bar/movement <i>speed</i> , not just load — moved explosively
Hypertrophy	67–85%	6–12	3–4	30–90 sec	Moderate load, moderate rest — metabolic stress is part of the stimulus
Muscular Endurance	<67%	12+ (often 15–25)	2–3	≤30 sec	Lighter load, minimal rest, higher total reps

Simple rule of thumb: as %1RM goes down, rep range goes up, and rest interval goes down. The four adaptations sit on one continuous spectrum, not four unrelated boxes.

5. Applying RIR and RPE

Two ways to gauge how hard a set actually was, used to autoregulate load session to session.

RPE (Rate of Perceived Exertion) — Borg CR10 scale, adapted for resistance training:

RPE	Meaning
10	Maximal effort — no more reps possible
9	Could do 1 more rep
8	Could do 2 more reps
7	Could do 3 more reps
≤6	Warm-up / light effort — 4+ reps left

RIR (Reps in Reserve) is just RPE inverted:

$$RIR = 10 - RPE$$

So RPE 8 = 2 RIR. RPE 9 = 1 RIR. RPE 10 = 0 RIR (true failure).

Why this matters in practice:

- A prescription of "4x8 @ RPE 8" tells the athlete/patient to pick a load where the last rep of each set leaves about 2 reps in the tank — the load auto-adjusts to a good day vs. a bad day, unlike a fixed weight target.
 - **Clinical/rehab use:** RIR is especially useful early in rehab when 1RM testing is inappropriate or unsafe. You can prescribe "3x12 @ RPE 6–7" without ever needing to know the patient's true max.
 - **General guideline for progressive overload:** once a patient/athlete can complete all prescribed sets at 2 RIR or lower (i.e., RPE 8+) with good form, that's your signal to increase load next session.
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6. Simple Ways to Progress and Regress an Exercise

You don't need a new exercise to make something harder or easier — these are the basic levers, and they're the same levers whether you're in a clinic or a weight room.

Ways to progress (make harder):

1. **Increase load** — more weight, same reps
2. **Increase reps or sets** — same load, more total volume
3. **Decrease base of support** — two feet → single leg; stable surface → unstable surface (foam pad, BOSU)
4. **Increase range of motion** — partial ROM → full ROM (e.g., box squat → full-depth squat)
5. **Increase speed/tempo demand** — slow controlled rep → explosive/plyometric rep
6. **Reduce points of external support** — hands-assisted → no assistance; both arms → single arm
7. **Add instability or perturbation** — stable load → unstable load (e.g., barbell → dumbbells → cables/bands)
8. **Combine movements** — single-joint → multi-joint, or add a second task (e.g., squat + overhead press)

Ways to regress (make easier) — same list, reversed:

1. **Decrease load**
2. **Decrease reps or sets**
3. **Increase base of support** — single leg → both feet; unstable → stable surface
4. **Reduce range of motion** — full squat → box squat/partial range
5. **Slow the tempo down** — remove explosive/ballistic demand, control every rep
6. **Add external support** — use a wall, rail, or assisted band
7. **Increase stability of the load** — cables/bands → dumbbells → barbell (more fixed path)
8. **Isolate the movement** — multi-joint → single-joint, remove secondary tasks

Practical framework: pick *one* lever at a time. Changing load and ROM and stability all in the same session makes it impossible to know which variable caused a problem if the patient/athlete struggles. This is the same principle whether you're advancing a post-op ACL patient or progressing an off-season squat.

7. Rest Intervals by Training Goal

Rest interval is dictated by which energy system fuels the set and how much recovery it needs before the next quality effort.

Goal	%1RM	Reps	Rest Interval	Rationale
Max Strength	85–100%	1–5	3–5 min	Full ATP-PC restoration needed for near-max force output
Power / Speed-Strength	75–90% (or lighter, max velocity)	1–5	2–5 min	Rate of force development degrades fast with fatigue — protect quality over density
Hypertrophy	67–85%	6–12	30–90 sec	Metabolic stress is part of the stimulus; shorter rest is tolerable
Muscular Endurance	Below 67%	12+	≤30 sec	Trains lactate buffering/tolerance; long rest defeats the purpose

ATP-PC recovery math (why strength/power rest is long): phosphocreatine stores replenish on a roughly exponential curve —

$$\% \text{ PCr Recovered} \approx 100 \times (1 - 0.5^{(t / 30)})$$

where t = rest time in seconds. At $t = 30\text{s} \rightarrow \sim 50\%$ recovered; $t = 90\text{s} \rightarrow \sim 87.5\%$; $t = 180\text{s}$ (3 min) $\rightarrow \sim 98\%$; $t = 300\text{s}$ (5 min) $\rightarrow \sim 99.7\%$. This is the actual physiological basis for the “3–5 min for max strength/power” guideline — you’re waiting for near-full ATP-PC restoration before asking for another max effort.

8. Force-Velocity Curve

The inverse relationship between the force a muscle can produce and the velocity of contraction:

$$F = F_0 - k \cdot v$$

where F_0 = theoretical max isometric force ($v = 0$), k = curve slope constant, v = contraction velocity. **Power = Force $\times$ Velocity**, which is why peak power occurs near the *middle* of the curve, not at either extreme — max force happens at $v \approx 0$ (isometric), and max velocity happens when load/force approaches 0 (unloaded jump/throw).

Where training methods sit on the curve:

Region	%1RM	Velocity	Example Method
Max Strength (high force, low velocity)	85–100%	Very slow	Heavy squat, deadlift
Strength-Speed	60–85%	Moderate	Dynamic-effort heavy loads (~1.0–1.5 m/s bar speed)
Speed-Strength	30–60%	Fast	Jump squats, med ball throws
Max Speed (low force, high velocity)	0–30% (bodyweight or unloaded)	Very fast	Sprinting, unloaded jumps

Clinical tie-in: an individual’s force-velocity profile (tested via loaded jump squats or sprint velocity decrements) tells you *which region* they’re weakest in — a “force-deficient” patient/athlete needs heavier work, a “velocity-deficient” one needs lighter/faster work. This is a step beyond generic rep prescriptions and shows individualized programming logic.

9. Training Each Energy System

Three systems, distinguished primarily by work:rest ratio and duration of effort:

System	Primary Fuel	Effort Duration	Work:Rest Ratio	Example Training Format
ATP-PC (Phosphagen)	Stored ATP/creatine phosphate	0–10 sec	1:12 to 1:20	10s sprint, then 2–3 min rest
Glycolytic (Anaerobic)	Muscle glycogen → lactate	10 sec – 2 min	1:3 to 1:5	30s shuttle run, then 90–150s rest
Oxidative (Aerobic)	Fat + glycogen (O ₂ -dependent)	2 min +	1:1 to 1:3 (or continuous)	Tempo runs, continuous conditioning

Worked example — building a glycolytic conditioning set: Work interval = 30 sec at high intensity. Using a 1:4 ratio → Rest = $30 \times 4 = 120 \text{ sec (2 min)}$. Repeat for prescribed number of reps/sets based on phase (higher volume off-season, lower volume/more sport-specific in-season).

Why the ratio matters more than the exact drill: get the work:rest ratio wrong (e.g., too little rest on glycolytic work) and you accidentally train a different system than intended — the athlete/patient shifts into aerobic pacing to survive the session rather than developing anaerobic power, which is a common conditioning mistake.

Season/rehab application:

- **Off-season / early rehab:** broadest development across all three systems, including aerobic base building (lower per-sport priority but still trains recovery capacity between efforts)
- **In-season / late-stage rehab:** conditioning narrows to sport-specific work:rest ratios that mimic actual competition demand (e.g., football ≈ ATP-PC/glycolytic dominant with 1:15+ ratios; soccer ≈ more oxidative-glycolytic blend with shorter rest)

Key Sources

1. NSCA — *Essentials of Strength Training and Conditioning*, 4th ed. (Human Kinetics)
2. Borg CR10 Scale — Borg, G. (1998). *Borg's Perceived Exertion and Pain Scales*. Human

Kinetics.

3. NSCA — periodization and program design principles (energy system training, force-velocity relationship)